



# Ad HOC Insights Consumer Goods



Created By Vidhi  
Sajnani

**AtliQ Hardwares is one of the leading computer hardware producers in India and has expanded well in other countries too.**

**However, the management noticed that they do not get enough insights to make quick and smart data-informed decisions. They want to expand their data analytics team by adding several junior data analysts. There are 10 ad hoc requests for which the business needs insights.**





1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

|   | market      |
|---|-------------|
| ▶ | India       |
|   | Indonesia   |
|   | Japan       |
|   | Philippines |
|   | South Korea |
|   | Australia   |
|   | Newzealand  |
|   | Bangladesh  |

2. What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields,

unique\_products\_2020  
unique\_products\_2021  
percentage\_chg

|   | unique_products_2020 | unique_products_2021 | percentage_chg |
|---|----------------------|----------------------|----------------|
| ► | 245                  | 334                  | 36.33          |

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,

segment  
product\_count

|   | segment     | product_count |
|---|-------------|---------------|
| ▶ | Notebook    | 129           |
|   | Accessories | 116           |
|   | Peripherals | 84            |
|   | Desktop     | 32            |
|   | Storage     | 27            |
|   | Networking  | 9             |



4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,

segment  
product\_count\_2020  
product\_count\_2021  
difference

|   | segment     | product_count_2020 | product_count_2021 | difference |
|---|-------------|--------------------|--------------------|------------|
| ▶ | Accessories | 69                 | 103                | 34         |
|   | Notebook    | 92                 | 108                | 16         |
|   | Peripherals | 59                 | 75                 | 16         |
|   | Desktop     | 7                  | 22                 | 15         |
|   | Storage     | 12                 | 17                 | 5          |
|   | Networking  | 6                  | 9                  | 3          |

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,

product\_code  
product  
manufacturing\_cost

|   | product_code | product                                   | manufacturing_cost |
|---|--------------|-------------------------------------------|--------------------|
| ▶ | A0118150101  | AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s ... | 240.5364           |
|   | A0118150101  | AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s ... | 0.8920             |

6. Generate a report that contains the top 5 customers who received an average high pre\_invoice\_discount\_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,

customer\_code  
customer  
average\_discount\_percentage

|   | customer_code | customer | average_discount_percentage |
|---|---------------|----------|-----------------------------|
| ► | 90002009      | Flipkart | 30.83                       |
|   | 90002006      | Viveks   | 30.38                       |
|   | 90002003      | Ezone    | 30.28                       |
|   | 90002002      | Croma    | 30.25                       |
|   | 90002016      | Amazon   | 29.33                       |



7. Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and make strategic decisions. The final report contains these columns:

Month  
Year  
Gross sales Amount

|   | Month     | fiscal_year | Gross_sales_Amount |
|---|-----------|-------------|--------------------|
| ▶ | November  | 2021        | 20464999.10        |
|   | October   | 2021        | 13218636.20        |
|   | December  | 2021        | 12944659.65        |
|   | January   | 2021        | 12399392.98        |
|   | September | 2021        | 12353509.79        |
|   | May       | 2021        | 12150225.01        |
|   | March     | 2021        | 12144061.25        |
|   | July      | 2021        | 12092346.32        |
|   | February  | 2021        | 10129735.57        |
|   | June      | 2021        | 9824521.01         |
|   | November  | 2020        | 7522892.56         |
|   | April     | 2021        | 7311999.95         |
|   | August    | 2021        | 7178707.59         |
|   | October   | 2020        | 5135902.35         |
|   | December  | 2020        | 4830404.73         |
|   | January   | 2020        | 4740600.16         |
|   | September | 2020        | 4496259.67         |
|   | February  | 2020        | 3996227.77         |
|   | August    | 2020        | 2786648.26         |
|   | July      | 2020        | 2551159.16         |
|   | June      | 2020        | 1695216.60         |
|   | May       | 2020        | 783813.42          |
|   | April     | 2020        | 395035.35          |
|   | March     | 2020        | 378770.97          |

8. In which quarter of 2020 did the maximum total\_sold\_quantity? The final output contains these fields sorted by the total\_sold\_quantity,

Quarter  
total\_sold\_quantity

|   | Quarter | total_sold_quantity |
|---|---------|---------------------|
| ▶ | Q1      | 7005619             |
|   | Q2      | 6649642             |
|   | Q4      | 5042541             |
|   | Q3      | 2075087             |

9. Which channel helped to bring more gross sales in the fiscal year 2021, and the percentage of contribution? The final output contains these fields,

channel  
gross\_sales\_mln  
percentage

|   | channel     | gross_sales_mln | percenatge |
|---|-------------|-----------------|------------|
| ▶ | Retailer    | 1219.08         | 73.23      |
|   | Direct      | 257.53          | 15.47      |
|   | Distributor | 188.03          | 11.30      |



10. Get the Top 3 products in each division that have a high total\_sold\_quantity in the fiscal\_year 2021? The final output contains these fields,

division  
product\_code  
product  
total\_sold\_quantity  
rank\_order

|   | division | product_code | product             | total_sold_quantity | rank_order |
|---|----------|--------------|---------------------|---------------------|------------|
| ▶ | N & S    | A6720160103  | AQ Pen Drive 2 IN 1 | 701373              | 1          |
|   | N & S    | A6818160202  | AQ Pen Drive DRC    | 688003              | 2          |
|   | N & S    | A6819160203  | AQ Pen Drive DRC    | 676245              | 3          |
|   | P & A    | A2319150302  | AQ Gamers Ms        | 428498              | 1          |
|   | P & A    | A2520150501  | AQ Maxima Ms        | 419865              | 2          |
|   | P & A    | A2520150504  | AQ Maxima Ms        | 419471              | 3          |
|   | PC       | A4218110202  | AQ Digit            | 17434               | 1          |
|   | PC       | A4319110306  | AQ Velocity         | 17280               | 2          |
|   | PC       | A4218110208  | AQ Digit            | 17275               | 3          |